TITLE IN ENGLISH 1234

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NASLOV V SLOVENŠČINI

Magistrsko delo

Supervisor: Prof. Name Surname1

Co-Supervisor: Prof. Name Surname2

Ljubljana, Slovenia, Month Year

To ...

Acknowledgments

Thank everyone who contributed to the thesis. Do not forget to mention any funding received.

Abstract

The English abstract should not take up more than one page.

Povzetek

Povzetek v slovenščini naj ne bo daljši od ene strani.

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Abbreviations

CPU ... central processing unit (a very long description demonstating line breaks in such cases)
JSI ... Jožef Stefan Institute
IPS ... International Postgraduate School

Symbols

$j^\star$ ... black-body irradiance
σ ... Stefan's (or Stefan-Boltzmann) constant

Glossary

Glossary of terms

Chapter 1

Introduction

1.1 Thesis Structure

The thesis should be structured as follows (note that while some parts are optional, their order is not):

- *Front Matter*
  - Title pages
  - Dedication (optional)
  - Acknowledgments
  - Abstract
  - Povzetek
  - Contents
  - List of Figures (required if the thesis contains figures)
  - List of Tables (required if the thesis contains tables)
  - List of Algorithms (required if the thesis contains algorithms)
  - Abbreviations (optional)
  - Symbols (optional)
  - Glossary (optional)
- *Main Matter*
  - 1 Introduction
  - $2, \ldots, n-1$ Chapters
  - n Conclusions

  The Introduction must clearly summarize the thesis from the topic application of the master thesis. The core text of the master thesis can be substituted by publications (or papers accepted for publication) in internationally recognized journals. In this case, the Introduction should clearly describe the scientific method and the candidate's contribution to any publication which has been produced by several authors. In the Discussion or Conclusions, the candidate should summarize coherently the results of his/her thesis.
- *Appendices* (optional)

  - Appendix A Title
  - Appendix B Title
  - ...

  If desired, the candidate can place some of his/her relevant papers in the appendices. In order to do so, he/she must obtain permissions from publishers.

- *Back Matter*
  - References
  - Bibliography
  - Biography
  - Index (optional)

In addition, each author must prepare the cover page and spine using the two `.doc` files provided by IPS.

1.2 Sectioning Example

This is a section. Some more text. Some more text. Some more text. Some more text. Some more text. Some more text. Some more text. Some more text. Some more text. Some more text. Some more text. Some more text.

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Chapter 2

Floating Bodies

Floating bodies are figures, tables and algorithms.

2.1 Figures

Captions should be placed below figures as shown in Figure 2.1. If a caption is shorter than the line width, it should be centered.

Figure 2.1: A large IPS logo.

On the other hand, if a caption is very long (see Figure 2.2), only its first (short) part should be put in the List of Figures.

Figure 2.2: A small IPS logo. The IPS has its own logo and a uniform graphic image, which is used on all its documents.

2.2 Tables

Similar rules apply also to captions of tables, with the exception that captions are placed above tables (see Table 2.1).

Table 2.1: A simple table.

A	B	C
12	9834	327
51	2234	97

2.3 Algorithms

Algorithm 2.1 presents an algorithm example.

Algorithm 2.1: An algorithm example.

```
Data: this text
Result: complete understanding
initialization;
while not at end of this document do
    read current section;
    if understood then
        go to next section;
        current section becomes this one;
    else
        go back to the beginning of current section;
    end
end
```

Chapter 3

Equations and Measurement Units

3.1 Equations

Small equations are often written in-line (within the text), for example $j^\star = \sigma T^4$, while larger ones need to be displayed in the following way:

$$\sigma = \frac{2\pi^5 k^4}{15c^2h^3} = 5.6704 \times 10^{-8} Js^{-1}m^{-2}K^{-4} \tag{3.1}$$

All displayed equations need to be numbered so that they can be referenced later in the text (for example, Eq. (3.1) presents the Stefan's (or Stefan-Boltzmann) constant).

3.2 Measurement Units

The candidate can choose a standard for measurement units and has to consistently use it throughout the thesis.

Chapter 4

Definitions and Theorems

4.1 Definitions

See the formal definition of the right triangle in Definition 4.1.

Definition 4.1 (Right triangle). A *right triangle* is a triangle in which one angle is a 90-degree angle.

4.2 Theorems

The Pythagorean theorem is a relation in Euclidean geometry among the three sides of a right triangle. It states that the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides [1].

Theorem 4.1 (Pythagorean theorem). *In every right triangle with sides a and b and hypotenuse c, the following holds:*

$$a^2 + b^2 = c^2 \tag{4.1}$$

See Appendix A for the proof of this theorem.

Chapter 5

Reference Formatting

References should be formatted using either the IEEE or APA 6th edition formatting style. This template uses the former one.

5.1 IEEE

Let us cite a journal article [2]. References to journal articles that have not yet been published should contain the `doi`, as in [3]. Multiple references can be cited at the same time [4], [5], [6], [7], [8]. Beside books and journal articles, parts of books [9], technical reports [10], PhD theses [11] and MSc theses [12] can be included in the references.

Finally, on-line sources can be referenced, too, see Section 4.2.

5.2 APA 6th Edition

Please see the template using the APA style.

Chapter 6

Conclusions

We came to the following conclusions ...

Appendix A

Proofs of Theorems

A.1 Proof of the Pythagorean Theorem

Let us prove the Pythagorean Theorem from page 9.

Proof. This proof is based on the proportionality of the sides of two similar triangles, that is, upon the fact that the ratio of any two corresponding sides of similar triangles is the same regardless of the size of the triangles.

Let ABC represent a right triangle, with the right angle located at C, as shown in Figure A.1. We draw the altitude from point C, and call H its intersection with the hypotenuse AB. Point H divides the length of the hypotenuse into two parts.

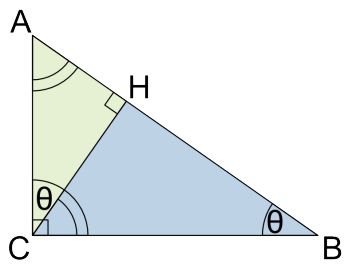

Figure A.1: Similar triangles used in the proof of the Pythagorean theorem.

The new triangle ACH is similar to triangle ABC, because they both have a right angle (by definition of the altitude), and they share the angle at A, meaning that the third angle will be the same in both triangles as well, marked as θ in Figure A.1. By a similar reasoning, the triangle CBH is also similar to ABC.

Similarity of the triangles leads to the equality of ratios of corresponding sides:

$$\frac{BC}{AB} = \frac{BH}{BC} \text{ and } \frac{AC}{AB} = \frac{AH}{AC}. \tag{A.1}$$

The first result equates $\cos\theta$ and the second result equates $\sin\theta$.

These ratios can be written as:

$$BC^2 = AB \times BH \text{ and } AC^2 = AB \times AH. \tag{A.2}$$

Summing these two equalities, we obtain:

$$BC^2 + AC^2 = AB \times BH + AB \times AH = AB \times (AH + BH) = AB^2, \tag{A.3}$$

which, tidying up, is the Pythagorean theorem:

$$BC^2 + AC^2 = AB^2. \tag{A.4}$$

□

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[3] T. Tušar and B. Filipič, "Visualization of Pareto front approximations in evolutionary multiobjective optimization: A critical review and the prosection method," *IEEE Transactions on Evolutionary Computation*, 2014, In press. DOI: 10.1109/TEVC.2014.2313407.

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Bibliography

Publications Related to the Thesis

All publications related to the thesis should be referenced in the text.

Journal Articles

M. Depolli, R. Trobec, and B. Filipič, "Asynchronous master-slave parallelization of differential evolution for multiobjective optimization," *Evolutionary Computation*, vol. 21, no. 2, pp. 261–291, 2013.

D. Mihailović, P. Kušar, J. Demšar, and V. V. Kabanov, "The determination of length scales associated with dynamic inhomogeneity in cuprate superconductors," *Journal of Superconductivity*, vol. 19, no. 1/2, pp. 43–47, 2006.

T. Tušar and B. Filipič, "Visualization of Pareto front approximations in evolutionary multiobjective optimization: A critical review and the prosection method," *IEEE Transactions on Evolutionary Computation*, 2014, In press. DOI: `10.1109/TEVC.2014.2313407`.

Conference Paper

M. Zupanc et al., "Removal of micropollutants by suspended biomass and biofilm processes and their improved removal by coupling biofilm process with alternative treatments," in *Proceedings of the 9th International Conference on Biofilm Reactors*, Paris, France: International Water Association, 2012.

Other Publications (optional)

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Biography

The author of this thesis ...

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